

# Physical and Electrolytic Properties of Partially Fluorinated Methyl Propyl Carbonate and Its Application to Lithium Batteries

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The physical and electrolytic properties, and charge-discharge characteristics for secondary lithium batteries of partially fluorinated methyl propyl carbonate (MPC) derivatives were examined. Dielectric constants of the fluorinated MPC derivatives except 2,2,3,3,3-pentafluoropropyl methyl carbonate (PFPMC) tend to increase with increasing the number of fluorine atom. Viscosity of PFPMC with the highest molar volume is lower than those of other fluorinated MPC derivatives. Donor numbers (DN) of the fluorinated MPC derivatives were estimated by <sup>29</sup>Si NMR chemical shifts of triphenylsilanol. DN of the fluorinated MPC derivatives except PFPMC tends to decrease with increasing the number of fluorine atom. 3-Fluoropropyl methyl carbonate (FPMC) shows the highest specific conductivity at the temperature above about 25 °C. Oxidative decomposition potentials of the fluorinated MPC derivatives tend to increase with increasing the number of fluorine atom. PC-FPMC electrolyte shows higher lithium electrode cycling efficiency. Every PC-based fluorinated MPC derivative electrolyte shows higher discharge capacities for Li | LiCoO<sub>2</sub> cells.

**Key Words :** Fluorinated Methyl Propyl Carbonate, Physical Properties, Donor Number, Lithium Batteries

## 1 Introduction

Secondary lithium batteries (lithium ion batteries) are becoming indispensable power sources for various portable electronic devices and electric vehicles. To develop the secondary lithium battery with better performance, many efforts to find improved solvents with oxidation durability, wide liquidus range and non-flammability have been made by many workers.<sup>1,2)</sup> Fluorinated organic solvents show very different physical properties such as dielectric constant, viscosity and polarity, compared with those of common organic solvents because of very high electronegativity, high ionic potential and low polarizability of fluorine atom.<sup>3,4)</sup> In particular, partially fluorinated organic solvents show very high polarity compared to perfluoro organic solvents. Partial fluorination is to be expected as an appropriate method of finding a solvent with good cell performance. We have investigated the physical and electrolytic properties of partially fluorinated cyclic and linear carbonates and their applications to lithium batteries.<sup>5-11)</sup>

We synthesized four kinds of partially fluorinated methyl propyl carbonate (MPC) derivatives such as 3-fluoropropyl methylcarbonate (FPMC), 3,3,3-trifluoropropyl methyl carbonate (TrFPMC), 2,2,3,3-tetrafluoropropyl methyl carbonate (TeFPMC) and 2,2,3,3,3-pentafluoropropyl methyl carbonate (PFPMC). MPC is a structural isomer of diethyl carbonate (DEC) as a typical and important cosolvent for lithium batteries. The purpose of the present paper is to elucidate the physical and electrolytic properties, and charge-discharge characteristics

for lithium batteries of the fluorinated MPC derivatives.

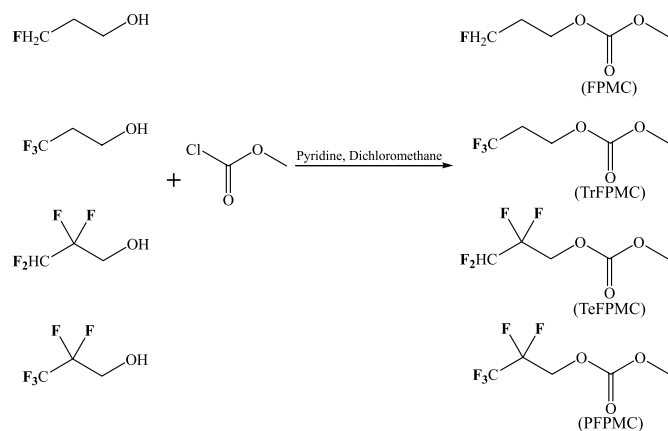
## 2 Experimental

### 2.1 Preparation of partially fluorinated MPC derivatives

Fluorinated MPC derivatives are synthesized by use of four kinds of fluoropropanol and methyl chlorocarbonate according to scheme 1. These fluorinated MPC derivatives were obtained in higher yields (80%). Confirmation of structures of the fluorinated MPC derivatives was carried out by GC-MS (JMS-SX102A) and <sup>1</sup>H-, <sup>13</sup>C-, <sup>19</sup>F-NMR (JEOL, JNM-LA500). The purities of these fluorinated MPC derivatives were measured as more than 99.9% by using GC (SHIMAZU, GC-1700).

### 2.2 Apparatus and measurements

The preparation of the solutions and the assembly of the cells were carried out in an argon-filled dry box. The apparatus and techniques for measurements of viscosity, dielectric constant and specific conductivity were reported previously.<sup>12-15)</sup> Linear potential sweep voltammetry (i-E curve) was performed using a computer-controlled electrochemical system (Hokuto Denko, Hz-3000). The <sup>29</sup>Si NMR spectra of triphenylsilanol for estimation of donor numbers of the fluorinated MPC derivatives were measured at 25 °C, referred to tetramethylsilane (TMS) as an external reference.<sup>16,17)</sup> The lithium electrode cycling efficiency (charge-discharge coulombic cycling efficiency of lithium electrode) was measured with a charge/discharge unit (HOKUTO DENKO, HJ-201B) and estimated by a galvanostatic plating-stripping reported



Scheme 1

by Koch and Brummer.<sup>18)</sup> A 2025-type coin cell made up by Li (anode), LiCoO<sub>2</sub> (cathode, 85 wt %)(PIOTREK), pioxel C100 (separator) was used for measurements of discharge capacity and cycling efficiency. The cathode also contained ketjenblack (7 wt %) and PVdF (8 wt %).

### 3 Results and Discussion

Figure 1 shows the temperature dependence of dielectric constants of the fluorinated MPC derivatives. The dielectric constant of every fluorinated MPC derivative becomes much higher than that of MPC. The dielectric constant of the fluorinated MPC derivatives tends to increase with increasing the number of fluorine atom. However, PFPMC with five fluorine atoms shows lower dielectric constant than that of FPMC. The introduction of five fluorine atoms into MPC seems to decrease the polarity compared to FPMC by steric hindrance or electron repulsion among fluorine atoms. Figure 2 shows the temperature dependence of viscosities of the fluorinated MPC derivatives. TeFPMC shows the highest viscosity at a whole range of temperature. In addition, the viscosity of PFPMC is lower than those of other fluorinated MPC derivatives. This is a similar behavior to that of the dielectric constants in Fig. 1. Arrhenius plots of  $\log \eta$  vs.  $1/T$  for fluorinated MPC derivatives yield straight lines, as shown in Fig. 3. From the slopes of the lines, the activation energies for the viscosity of the fluorinated MPC derivatives are found to be 20.0, 16.3, 15.5, 14.5 and 11.7 kJ mol<sup>-1</sup> for TeFPMC, TrFPMC, FPMC, PFPMC and MPC, respectively. PFPMC shows the lowest activation energy among the fluorinated MPC derivatives. Accordingly, PFPMC with five fluorine atoms seems to have very lower molecular interaction compared to other fluorinated MPC derivatives. Figure 4 shows the temperature dependence of molar volumes of the fluorinated MPC derivatives. It seems that the order of molar volumes except FPMC is dependent on the molecular weight of the fluorinated MPC derivatives. In addition, PFPMC with the highest molar volume and molecular weight shows the lowest viscosity among the fluorinated MPC derivatives in Fig. 2. This fact means also that PFPMC has very lower molecular interaction. On the other hand, higher viscosity of FPMC with lower molar volume compared with that of MPC seems to be depen-

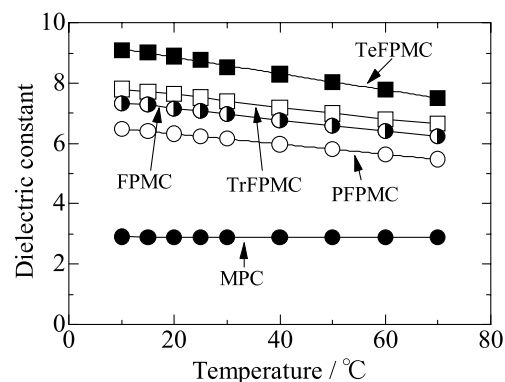


Fig. 1 Temperature dependence of dielectric constants of fluorinated MPC derivatives.

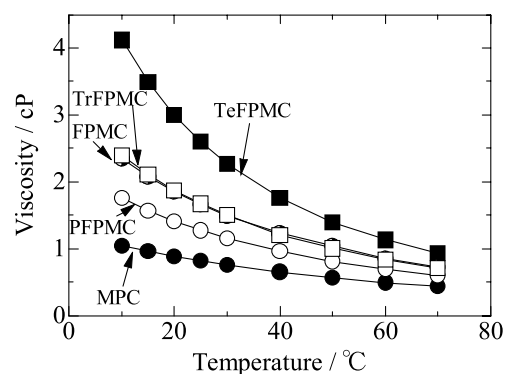


Fig. 2 Temperature dependence of viscosities of fluorinated MPC derivatives.

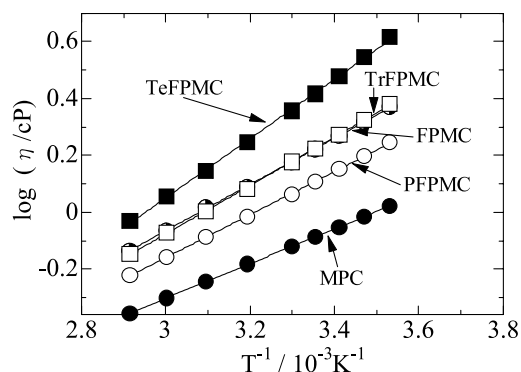


Fig. 3 Plots of the logarithm of viscosity vs.  $1/T$  for fluorinated MPC derivatives.

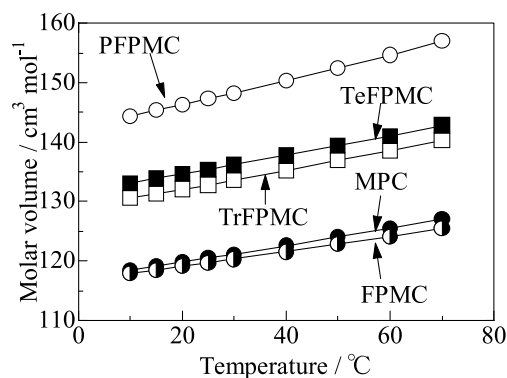


Fig. 4 Temperature dependence of molar volumes of fluorinated MPC derivatives.

dent on higher molecular interaction.

From the point of view of the microscopic properties, it is well known that the donor number (donicity, DN) of solvents is easily estimated by NMR spectroscopic measurements.<sup>16,19)</sup> The measurement of the  $^{29}\text{Si}$  NMR chemical shift is an appropriate means of obtaining the donor numbers of solvents. We have already examined the relationship between the  $^{29}\text{Si}$  NMR chemical shifts of triphenylsilanol and the donor numbers of various solvents.<sup>20)</sup> This method gave a fairly good calibration curve. We obtained the donor numbers of the fluorinated MPC derivatives, as can be seen from Table 1. The donor numbers of the fluorinated MPC derivatives generally tend to decrease with increasing the number of fluorine atom. However, donor number of PFPMC becomes higher rather than that of TeFPMC. This means that solvation of triphenylsilanol to PFPMC compared to TeFPMC is high due to the steric hindrance or electron repulsion among fluorine atoms as described previously.

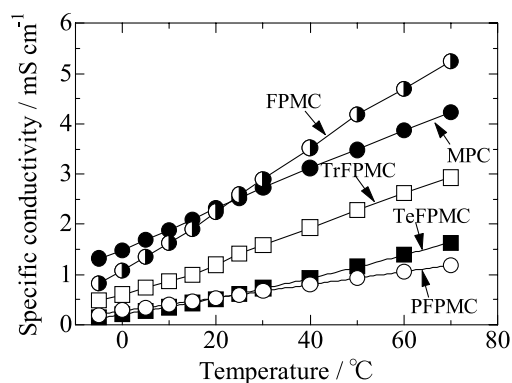
Figure 5 shows the temperature dependence of specific conductivities of the fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$ . At about  $25^\circ\text{C}$ , the order of the conductivities of MPC and FPMC or TeFPMC and PFPMC changes. Therefore, the conductivity of the solvent with higher donor number tends to become higher at the temperature below about  $25^\circ\text{C}$ . It is thought that the difference of solvation in these solvents affects the specific conductivity. On the other hand, the conductivity at about above  $25^\circ\text{C}$  mainly seems to be dependent on the viscosity rather than dielectric constant of the solvents. Anyhow, the conduction mechanisms of these electrolytes are complicated.

Figure 6 shows  $i$ - $E$  curves in the fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$ . Oxidative decomposition potential of the fluorinated MPC derivatives tends to increase with increasing the number of fluorine atom with high electron withdrawing. The introduction of many fluorine atoms into MPC is effective for the increase of the oxidation durability.

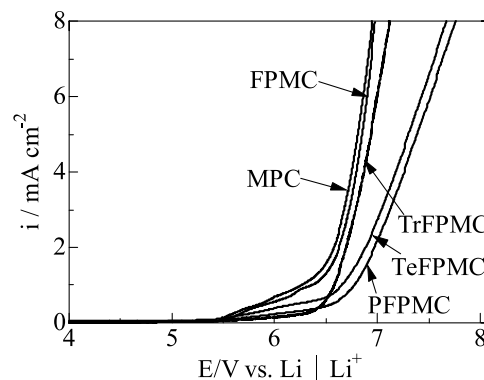
Figure 7 shows the lithium electrode cycling efficiencies in PC and PC-based fluorinated MPC derivative solutions. It is difficult to obtain the cycling efficiencies in single fluorinated MPC derivative solutions because the discharge curves are rarely measured in these single solutions. PC-FPMC solution shows higher cycling efficiency compared to PC and other PC-based fluorinated MPC derivative solutions. The charge-discharge process, that is, deposition-dissolution process of lithium on the electrode is smoothly occurred in the PC-FPMC solution. Figure 8 shows the SEM photographs on Ni electrode in PC and PC-based fluorinated MPC derivative solutions. In general, the lithium electrode cycling efficiency depends primarily on the morphology, chemical composition and thickness of the films formed on the electrode. The films on the electrode in PC and PC-MPC solutions are appreciably different from those in PC-based fluorinated MPC derivative solutions. In particular, a lithium dendrite appears to be formed in PC-MPC solution. On the other hand, it seems that a film in PC-FPMC solution with higher cycling efficiency in Fig. 7 is relatively

**Table 1** Donor numbers (DN) and  $^{29}\text{Si}$  NMR chemical shifts of triphenylsilanol in the fluorinated MPC derivatives.

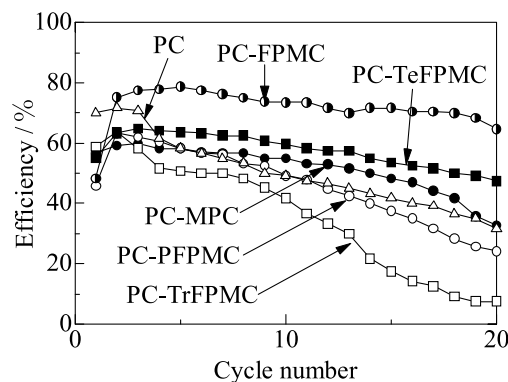
| Solvent | Chemical shift/ppm | DN |
|---------|--------------------|----|
| MPC     | -15.26             | 12 |
| FPMC    | -15.11             | 11 |
| TrFPMC  | -14.87             | 9  |
| TeFPMC  | -14.19             | 4  |
| PFPMC   | -14.45             | 6  |



**Fig. 5** Temperature dependence of specific conductivities of fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$ .



**Fig. 6**  $i$ - $E$  curves in fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$  using a platinum working electrode at a scan rate of  $5 \text{ mV s}^{-1}$  at  $25^\circ\text{C}$ .



**Fig. 7** Variation of lithium electrode cycling efficiencies in PC and PC-based (mol ratio 1 : 1) fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$  at  $25^\circ\text{C}$ .

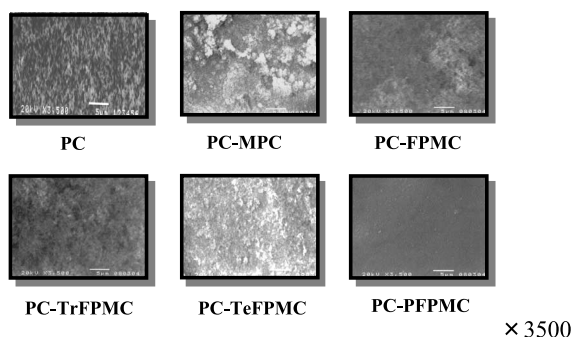


Fig. 8 SEM photographs on Ni electrode in PC and PC-based (mol ratio 1 : 1) fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$  after discharged of 10th cycle.

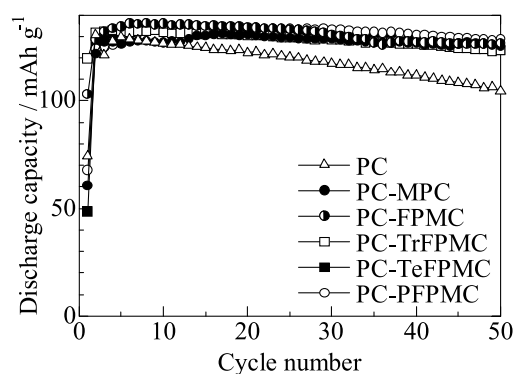


Fig. 9 Discharge capacities for Li | LiCoO<sub>2</sub> 2025 coin cells in PC and PC-based (mol ratio 1 : 1) fluorinated MPC derivative solutions containing  $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$  at 25 °C. Charge ; 0.2C CCCV at 4.2 V, cut off at 5 h  
Discharge ; 0.2C CC, cut off at 3.0 V

homogeneous and uniform compared to other PC-based fluorinated MPC derivative solutions.

Figure 9 shows the discharge capacities for Li | LiCoO<sub>2</sub> cells in PC and PC-based fluorinated MPC derivative solutions. The capacity in every PC-based solution is higher than that in PC solution. The difference of the discharge capacities for each PC-based fluorinated MPC derivative solution is very small. In addition, each PC-based solution has the discharge capacities more than 120 mAh g<sup>-1</sup> at 50th cycle number.

#### 4 Conclusions

Dielectric constant of the fluorinated MPC derivatives tends to increase with increasing the number of fluorine atom. However, PFPMC with five fluorine atoms shows lower dielectric constant than that of FPMC with one fluorine atom. TeFPMC with four fluorine atoms shows the highest viscosity. Viscosity of PFPMC with the highest molar volume is lower than those of other fluorinated MPC derivatives. Donor number (DN) of the fluorinated MPC derivatives tends to decrease with increasing the

number of fluorine atom. However, DN of PFPMC becomes higher rather than of TeFPMC. The order of the specific conductivities of MPC and FPMC or TeFPMC and PFPMC changes at about 25 °C. The conductivity of the solvent with higher DN tends to become higher at the temperature below about 25 °C. The conductivity at about above 25 °C is mainly dependent on the viscosity of the solvents. Oxidative decomposition potential of the fluorinated MPC derivatives tends to increase with increasing the number of fluorine atom. PC-FPMC solution shows higher lithium electrode cycling efficiency. The Li | LiCoO<sub>2</sub> cells in PC-based fluorinated MPC derivatives solutions show higher discharge capacities more than 120 mAh g<sup>-1</sup> at 50th cycle number. The capacities are almost independent of the number of fluorine atom.

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